FINAL PRESENTATION · SESSION 2026–27

AI-Based Real-Time Object Detection and Perception System for Autonomous Vehicles

YOLO11n + BYTETRACK + TTC / ROI + FASTAPI DASHBOARD

Camera-only Mini Project: detect, track, measure distance, fit a dynamic corridor, compute TTC, publish an advisory CONTINUE / WARNING / BRAKE — and never invent ego speed or fake lanes.

VIDEO → YOLO11 → TRACK → DISTANCE → ROI → TTC → DASH

0.418	26.4	10	16
MAP50 · BDD VAL	FPS GPU · RTX 2050	DRIVING CLASSES	TABULATED TESTS

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IPS Academy, IES, Indore · 2026–27

Pixels are not perception

Urban driving is an open-world visual problem. A box on an image is not enough: a car in the oncoming lane is not the same hazard as a car two metres ahead in the ego path.

SEVEN QUESTIONS EVERY FRAME MUST ANSWER

1. **What** objects are present — class + confidence.
2. **Where** they are — box in the image plane.
3. **Which** of them lie in the driving corridor.
4. **How** they move — identity across frames.
5. **How close** — monocular D = (H · F) / h.
6. **Approaching?** — closing speed and TTC.
7. **Dangerous?** — rule table → CONTINUE / WARNING / BRAKE.

ITEM	IN SCOPE	OUT OF SCOPE
Input	One video or camera	LiDAR, radar, multi-cam
Detection	YOLO11 COCO / BDD	Two-stage / transformers
Tracking	ByteTrack / BoT-SORT	Trained Re-ID
Geometry	Pinhole D + classical ROI	Metric 3D cuboids
Safety	Advisory TTC rules	Actuation, ISO 26262
Software	FastAPI + static UI	Cloud fleet / mobile app

Honesty constraint: if ego speed is not sensed it is published as null. Poor lane evidence is labelled uncertain — never faked.

EIGHT DELIVERED OBJECTIVES

1. Real-time detection of road objects (YOLO11n).
2. Identity tracking across frames (ByteTrack).
3. Corridor perception — in / near / outside / uncertain.
4. Relative motion from consecutive frames.
5. Monocular distance from box height.
6. TTC + four-level risk (SAFE…CRITICAL).
7. Recorded video and live camera input.
8. Nine-page dashboard of every measured number.

Twelve sources, one composition

Four clusters: single-stage detection, driving datasets, multi-object tracking, 3D surveys — plus classical edges, Hough, pinhole geometry and TTC. We do not claim a new detector.

#	SOURCE	APPROACH	KEY RESULT	USE IN AUTOSENSE
1	Redmon et al., CVPR 2016	YOLO one-shot CNN	63.4% mAP · 45 FPS	Real-time backbone
2	Yu et al., CVPR 2020	BDD100K benchmark	24.5 AP · 56.7 MOTA	10-class driving data
3	Zhang et al., ECCV 2022	BYTE association	69.6 MOTA · 71.3 IDF1	Default tracker
4	Mao et al., 3D survey	LiDAR / camera / fusion	BtcDet 78.58% L1	Why we stay 2D
5	Qian et al., PR 2022	Taxonomy of 3D	AP3D / NDS	Depth / fusion gap
6	Bewley et al., SORT	Kalman + Hungarian	Realtime MOT	Tracking-by-detection
7	Wojke et al., DeepSORT	Appearance Re-ID	Fewer ID switches	Identity stability
8	Aharon et al., BoT-SORT	Bag-of-tricks MOT	Strong IDF1	Optional UI tracker
9	Canny, 1986	Optimal edge detector	Standard operator	Dynamic ROI stage
10	Duda & Hart, Hough	Line parameter voting	Robust segments	Lane candidates
11	Ultralytics YOLO11	Train / track / export	YOLO11n 2.59M	Runtime API
12	Lin et al., COCO	80-class protocol	Default names	Bundled yolo11n.pt

> Gap: an inspectable real-time student stack that (i) runs on one camera, (ii) exposes every intermediate, (iii) states uncertainty, (iv) never fabricates ego speed. Composition is the claim.

03 · DATASET, PLATFORM & STACK

BDD100K official split, laptop GPU runtime

70,000
TRAIN IMAGES

10,000
VAL IMAGES

20,000
TEST IMAGES

1.29M
TRAIN OBJECTS · 0 INVALID

SPLIT	IMAGES	LABELS	OBJECTS	INVALID
Train	70 000	70 000	1 288 405	0
Validation	10 000	10 000	185 578	0
Test	20 000	20 000	367 728	0

TEN CLASSES · CARS DOMINATE

person · rider · car · bus · truck · bike · motor · traffic light · traffic sign · train · 15 val



TRAIN HOST

Kaggle Tesla T4 · 20 epochs · batch 16 · AMP · imgsz 640 · patience 6 · mosaic closed near end · HSV, ±3°, translate 0.1, scale 0.5, flip 0.5.

RUNTIME HOST · 27 SEP 2026

Windows i5-13420H / RTX 2050 · 540p dashcam @640. GPU 26.4 fps / 30.9 ms. CPU 16.2 fps / 54.4 ms. Boot → first frame 15.7 s.

STACK

Python 3.11 · Ultralytics 8.4 · PyTorch CUDA · OpenCV · NumPy · FastAPI · uvicorn · vanilla HTML/CSS/JS · pytest. Repo **Sarthak156/AutoSense_DR**. Notebook **autonomous-perception**.

04 · IMPLEMENTATION · SEVEN MODULES

One learned model. Geometry and rules for the rest.

VIDEO — YOLO11 — TRACK — DISTANCE — ROI — TTC — DASH

1 · Input
VideoCapture or camera index. Routes /api/source/video|camera. Play / pause / resume / restart / stop. Idle frame: NO INPUT SOURCE.

2 · Detect
YOLO11n track/predict · conf 0.35 · IoU 0.45 · imgsz 640. 182 layers · 2 591 790 params. Swap COCO 80-class or BDD 10-class .pt live.

3 · Track
persist=True · bytetrack.yaml default · botsort.yaml optional. Registry keyed by ID. Disable tracking → TTC unavailable (correct).

4 · Dynamic ROI
Gray → 5×5 blur → Canny → HoughLinesP → RANSAC / quadratic. Reuse ≤ 8 with decay, then fallback trapezoid. Never drops a box.

5 · Distance
D = (H · F) / h · F default 700 px (live-editable). Smooth window 5. Heights: person 1.70 · car 1.50 · truck/bus 3.20 · train 3.50 m.

6 · TTC + Risk + Serve
Closing speed from 8-sample deque. TTC cap 60 s. CRITICAL D≤3 m / 1 s → BRAKE. FastAPI MJPEG + WS ~5 Hz · nine pages.

ROIObjectAssociator returns in / near / outside / uncertain. Safety uses boolean **in_path** only. Overlay HUD draws corridor, boxes, IDs, distance. JPEG quality 80. Config patches deep-merge under a lock.

05 · MODEL TRAINING & EVALUATION

YOLO11n · 20 epochs · val on 10 000 images

0.418
MAP50 · ALL

0.227
MAP50–95

0.651
PRECISION

0.386
RECALL

0.721
CAR MAP50

CLASS	IMAGES	INST.	P	R	MAP50	MAP50-95
all	10 000	185 578	0.651	0.386	0.418	0.227
person	3 220	13 265	0.640	0.433	0.488	0.225
rider	515	649	0.573	0.290	0.312	0.143
car	9 879	102 540	0.702	0.685	0.721	0.434
bus	1 242	1 597	0.623	0.453	0.512	0.386
truck	2 689	4 247	0.633	0.482	0.534	0.379
bike	578	1 007	0.482	0.282	0.290	0.137
motor	334	452	0.620	0.220	0.286	0.132
traffic light	5 653	26 891	0.578	0.529	0.505	0.173
traffic sign	8 221	34 915	0.656	0.491	0.536	0.266
train	14	15	1.000	0.000	0.000	0.000

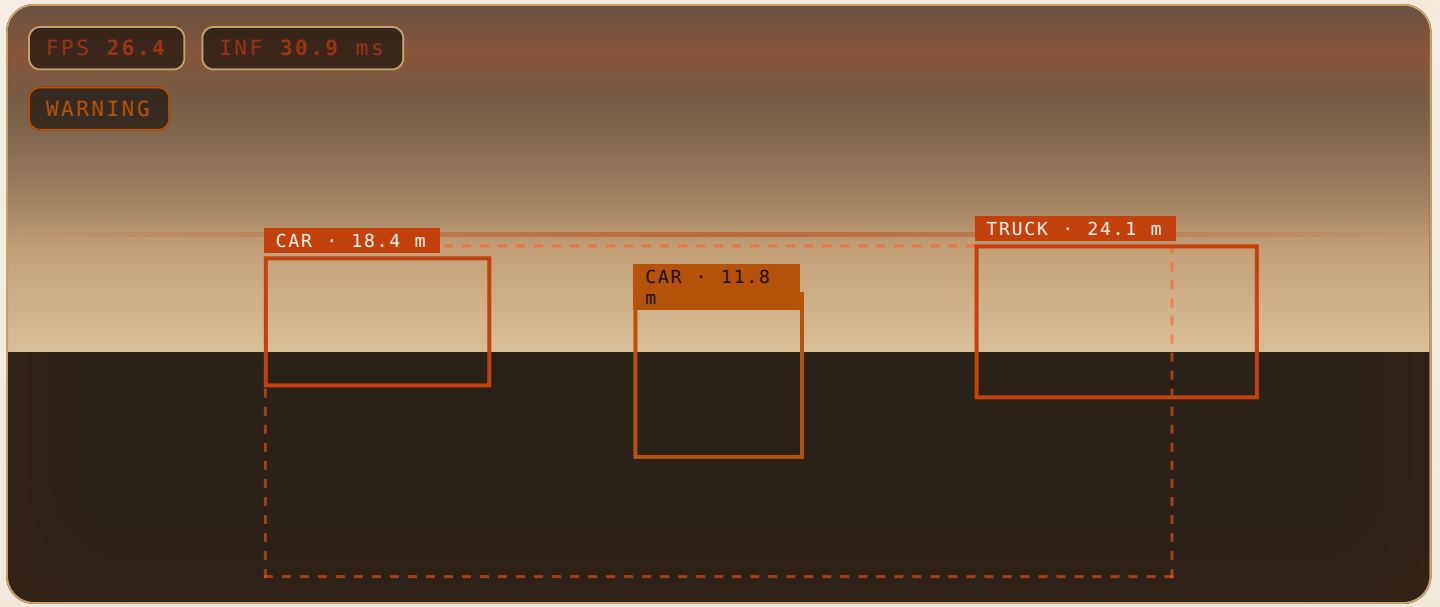
TRAINING RECIPE

COCO-pretrained YOLO11n → BDD100K. 20 epochs · batch 16 · AMP · patience 6 · 182 layers · 2.59M params (2.58M fused). T4: 0.1 ms pre · **1.3 ms inf** · 1.5 ms post. Exported **best.pt** + ONNX 10.1 MB.

HONEST IMBALANCE

102 540 car instances vs 15 train instances. Cars carry the score. Train mAP50 = 0 — reported, not hidden. Extra notebook val/test cells failed on a path mismatch; Table 5.5 is the trainer’s own val of best.pt at epoch 20.

Measured on an RTX 2050 — not a slide mock



26.4
FPS GPU 540P

16.2
FPS CPU

30.9
MS GPU INF

54.4
MS CPU INF

CHECK	OBSERVED
GET /healthz	ok true · state running
GET /api/models	yolo11n.pt · 15.9 MB · 80 cls
GET /api/video/frame	JPEG magic ffd8
WS /ws	fps 24.3 on connect
GET /api/report ~9 s	22 frames · 121 det · 23.7 fps
device cpu → auto	16.2 fps ↔ 26.4 fps

Nine pages: Dashboard, Video Input, Detection, Tracking, Distance & ROI, Safety, Visualizations, Model, Settings. Cold sweeps 8–11 fps (GPU power-state). 1080p upload ≈ 7 fps. Contended GPU can collapse inference — benchmark idle.

07 · SAFETY LAYER · RULE TABLE + TRACES

In-path gating, not raw TTC

CONTINUE
SAFE · not in-path

WARNING
D ≤ 12 m or TTC ≤ 4 s

BRAKE
CRITICAL / DANGER · advisory

LEVEL	DISTANCE	TTC	DECISION
CRITICAL	D ≤ 3.0 m	TTC ≤ 1.0 s	BRAKE
DANGER	D ≤ 6.0 m	TTC ≤ 2.0 s	BRAKE
WARNING	D ≤ 12.0 m	TTC ≤ 4.0 s	WARNING
SAFE	else / not in-path	—	CONTINUE

CLASS	ASSUMED H	CLASS	ASSUMED H
person / rider	1.70 m	car	1.50 m
truck / bus	3.20 m	motorcycle	1.40 m
bicycle	1.50 m	train	3.50 m
traffic light	0.60 m	traffic sign	0.75 m

TRACE A · tunnel_drive.mp4 · imgsz 640

WARNING / WARNING

Closest in-path **11.78 m** (≤ 12 m warning). Min TTC **4.63 s** (above 4 s TTC trigger). Distance rule fired. Not simulated.

TRACE B · dashcam_city.mp4

SAFE / CONTINUE

Min TTC **0.91 s** on an object **outside** the corridor. In-path gating refused BRAKE. ROI never dropped the detection.

Order: not in_path → SAFE; missing D → SAFE; else thresholds. TTC None if closing speed ≤ 0.1 m/s. Advisory only — must not be wired to a vehicle.

V-model + 16 cases — then harden

TC-01 … TC-16 · ALL PASS

TC-01	Curved lane quadratic fit
TC-02	Multi-lane centre-prior pair
TC-03	Search centre ≠ static trap
TC-04	Missing markings → fallback
TC-05	Hard shadows uncertain
TC-06	Crosswalk bars rejected
TC-07	Aspect jump clears history
TC-08	Reuse ≤ 8, then fallback
TC-09	Retain outside boxes
TC-10	Overlap association
TC-11	/healthz running
TC-12	JPEG ffd8 stream
TC-13	GPU 26.4 fps / 30.9 ms
TC-14	CPU 16.2 fps / 54.4 ms
TC-15	In-path WARNING 11.78 m
TC-16	Out-of-path TTC 0.91 s SAFE

pytest without weights (ROI + association + render layers). No k-fold — official BDD split is the substitute. Manual: pause/resume, loop-on-EOF, CAMERA ERROR, MODEL RUNTIME UNAVAILABLE, WS → 350 ms poll.

LIMITS

Inherent

Pinhole D is not LiDAR. Classical ROI fails at night, rain, worn paint, intersections. Risk ignores class and ego speed (null). Train class unlearnable. No auth — localhost only.

NEXT

Harden

JSONL logs, incident stills, optional ego-speed, F calibration wizard, per-class risk, TestClient smokes, ONNX / TensorRT, Dockerfile, config persistence.

NEVER

Non-goals

No actuation. No fake speed. No ADAS / ISO 26262 claim. No stereo/LiDAR in this Mini Project. WARNING/BRAKE are UI labels.

09 · CONCLUSION

From detection to understanding

The Mini Project built a camera-based real-time perception system for research, not a self-driving stack. Existing methods, composed, measured, and discussed without exaggeration.

0.418	0.721	26.4	11.78 m
MAP50 · 10K VAL	CAR MAP50	FPS · RTX 2050	IN-PATH WARNING

WHAT SHIPPED

YOLO11 detect + ByteTrack/BoT-SORT + pinhole distance + training-free corridor + TTC + advisory risk + nine-page FastAPI dashboard. Every FPS, inference_ms, distance and risk label is traceable. Unavailable signals stay unavailable.

SEES → TRACKS → MEASURES → ADVISES

TEAM · CSE (DATA SCIENCE) · 2026–27

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Questions?

THANK YOU · AUTOSENSE · COMPOSITION IS THE CONTRIBUTION